

Moligo Technologies – Investment Memo

1. Company Snapshot

Company Information

Item	Details
Company	Moligo Technologies
Location	Solna, Sweden
Founded	2019
Status	Private
Employees	~16
Therapeutic Focus	Gene therapy, gene editing, synthetic biology
Core Platform	Enzymatic synthesis of long, ultrapure single-stranded DNA (ssDNA)
Lead Applications	Gene editing donor templates, cell therapy, diagnostics, DNA nanotechnology

Company Overview

Moligo is a biotechnology company founded by researchers from Karolinska Institutet. The company has developed a proprietary enzymatic DNA synthesis platform capable of producing long, highly pure single-stranded DNA molecules at industrial scale.

The company aims to improve the safety, efficacy and affordability of gene-based technologies and therapies by overcoming limitations associated with traditional phosphoramidite chemistry and viral-vector-based approaches.

Funding Overview

- Seed/pre-seed stage
- Funding estimates:
 - Crunchbase: ~\$170K
 - PitchBook: ~\$782K
- Investors include:
 - Colibri Ventures
 - Vento
 - Nordic Science Capital
 - EIT Health
 - Phase2Phase Biopharma

Key observation: Moligo remains significantly less capitalized than leading competitors despite generating notable scientific validation.

2. Core Problem

Current Limitations of DNA Manufacturing

The dominant method for synthetic DNA production remains phosphoramidite chemistry. While highly effective for short oligonucleotides, this approach faces several limitations:

- Practical length limitation of approximately 200–300 bases
- Accumulation of truncated products
- Reduced fidelity with increasing sequence length
- Challenges with repetitive or highly structured sequences
- Increasing production costs for longer constructs

Why This Matters

Demand for long, highly pure DNA is increasing across:

- Gene editing
- Cell therapies
- CRISPR-based applications
- Synthetic biology
- DNA nanotechnology
- Advanced diagnostics

Many next-generation therapies require DNA constructs that exceed the practical limits of conventional synthesis methods.

3. Technology & Scientific Differentiation

Platform

Moligo has developed a proprietary Enzymatic Injection Molding Technology that enables:

- PCR-free production
- Long ssDNA (>10 kb)
- High sequence fidelity
- Industrial-scale manufacturing
- Production of both linear and circular ssDNA
- Incorporation of modified nucleotides

Claimed Performance

- Up to 10 kb single-stranded DNA
- 99.9999% fidelity
- 100% sequence verification
- Production batches approaching gram scale

Scientific Validation

In November 2025, Moligo and Collectis published results in Nature Communications demonstrating efficient gene insertion using circular ssDNA donor templates in hematopoietic stem cells.

Key findings included:

- 3–5× higher knock-in efficiency compared with linear ssDNA
- Improved engraftment relative to AAV6 approaches
- Potential applicability to HSPCs and primary T cells

These results provide proof-of-concept support for non-viral gene insertion strategies.

Why It Could Work

- Addresses known limitations of viral vectors
- Supports insertion of large genetic payloads
- Reduced cytotoxicity relative to some existing approaches
- Compatible with emerging gene-editing workflows

Why It Might Fail

- Dependence on production of an initial high-fidelity template
- Scalability remains largely unproven commercially
- Limited public evidence regarding manufacturing economics
- Lack of GMP manufacturing infrastructure

4. Commercial Applications

Application	Relevance
Gene Therapy	Large donor templates for gene insertion
CRISPR HDR	Long repair templates
CAR-T / Cell Therapy	Lower toxicity gene insertion
Diagnostics	High-purity reference standards
DNA Origami	Custom scaffold production

Application	Relevance
Drug Delivery Nanostructures	Programmable DNA nanocarriers
mRNA Manufacturing	IVT templates
Gene Synthesis & Cloning	Large construct generation

5. Intellectual Property & Moat

Proprietary Platform

Moligo's primary moat appears to derive from:

- Enzymatic Injection Molding Technology
- Manufacturing know-how
- Production of long ultrapure ssDNA
- Ability to produce both circular and linear formats

Open Questions

- Scope and breadth of patent protection
- Freedom-to-operate versus larger competitors
- Defensibility of manufacturing process
- Potential barriers to replication

Key Question for Investors

Is Moligo's competitive advantage primarily patent-driven, process-driven, or execution-driven?

6. Market & Competitive Landscape

Industry Overview

The field is increasingly moving toward:

- Non-viral gene delivery
- Long DNA donor templates
- Circular ssDNA architectures
- Industrial-scale DNA manufacturing

Competitor Comparison

Company	Technology	Main Advantage	Main Risk
Touchlight	Circular ssDNA engineering	GMP maturity, pharma partnerships	Regulatory precedent still developing

Company	Technology	Main Advantage	Main Risk
Ansa Biotechnologies	Enzymatic DNA synthesis	Funding and sequence length leadership	Commercial adoption still developing
Kano Therapeutics	Circular ssDNA templates	Therapeutic focus	Early-stage execution risk
Full Circles Therapeutics	Circular ssDNA engineering	Strong scientific momentum	Early clinical validation
GXstrands / CPTx	Fermentation-based ssDNA	Potential cost advantage	Less mature platform

Competitive Position

Moligo appears strongest in:

- Scientific credibility
- Academic origins
- European ecosystem positioning
- Long, ultrapure ssDNA production

Moligo appears weakest in:

- Capitalization
- Manufacturing scale
- Commercial infrastructure

7. Key Risks

Technical Risk

- Manufacturing complexity
- Template fidelity dependence
- Reproducibility at industrial scale

Regulatory Risk

- Limited regulatory precedent for non-viral DNA approaches
- Long-term safety profile remains under evaluation

Financing Risk

- Funding significantly below leading competitors
- Future scale-up likely requires substantial capital

Manufacturing Risk

- GMP capability not yet established

- Commercial-scale economics remain uncertain

Competitive Risk

- Well-funded US competitors
- Rapid technological evolution
- Potential IP challenges

8. SWOT Analysis

Strengths Differentiated enzymatic platform Strong scientific validation Strategic partnerships Linear and circular ssDNA capability	Weaknesses Limited funding No GMP manufacturing Small commercial team Length ceiling relative to some competitors
Opportunities Growth of non-viral gene editing CDMO positioning Large-gene therapy applications Increasing demand for ssDNA	Threats Better-funded competitors IP disputes Delayed fundraising Rapid technology shifts

9. Key Catalysts

Near-Term

- Series A financing
- New strategic partnerships
- Additional validation studies
- GMP manufacturing progress

Medium-Term

- Clinical adoption of ssDNA donor templates
- Expansion into pharmaceutical manufacturing
- Regulatory milestones in non-viral gene editing

10. Investment Thesis

Bull Case

- Scientifically differentiated platform
- Strong external validation despite limited funding
- Positioned within a rapidly growing segment of gene editing infrastructure
- Multiple commercial applications beyond therapeutics

Bear Case

- Significant funding disadvantage versus competitors
- Manufacturing scale remains unproven
- Commercial adoption timeline uncertain
- Competitive landscape becoming crowded

Conclusion

Moligo has established a strong scientific position within the emerging long-ssDNA manufacturing ecosystem. The company's most compelling characteristic is the apparent disconnect between its scientific output and its modest capital base — a dynamic that cuts both ways: it suggests either significant upside if funding follows, or meaningful execution risk if it doesn't.

The central investment question is whether Moligo can translate scientific leadership into manufacturing scale and commercial adoption before larger, better-funded competitors establish dominant positions. Concretely, a credible Series A milestone would likely require at least one of: initiation of GMP-certified manufacturing, a signed supply or co-development agreement with a pharmaceutical partner, or replication of the Cellectis results in a second independent therapeutic context. Without progress on one of these fronts within 18–24 months, the risk of being outpaced by better-capitalized US competitors becomes material.

If Moligo achieves these milestones, it could become a critical enabling platform for non-viral gene editing and next-generation genetic medicines, a picks-and-shovels position in one of the most capital-intensive areas of modern biotechnology.